

PROFESSIONAL SUMMARY

AI/ML Engineer with 4 years of experience designing production-grade Generative AI applications using **RAG**, GraphRAG, Agentic workflows, and hybrid retrieval systems. Experienced in building end-to-end AI pipelines—from large-scale data ingestion and processing to retrieval, evaluation, and deployment—using local and cloud-based LLMs. Strong foundation in **machine learning**, **deep learning**, and **statistics**, with a focus on building scalable, cost-efficient, and maintainable AI solutions.

TECHNICAL SKILLS

Languages: Python, SQL, R

GenAI / LLMs: RAG, GraphRAG, Agentic Workflows, Prompt Engineering, LoRA Fine-tuning, Local LLM Inference, RAGAS

Frameworks & Libraries: LangChain, LangGraph, PyTorch, TensorFlow, Scikit-learn, Pandas, Selenium, BERTopic, Pydantic, MCP

Retrieval & Vector Search: Qdrant, Hybrid Search (Dense + BM25), HNSW Indexing, Cross-Encoder Reranking, Sentence-Transformers

ML / NLP: Regression, Clustering, PCA, Topic Modeling, BERT Embeddings, HDBSCAN

Data / MLOps / Infra: Docker, MLflow, DVC, Airflow, Azure Data Factory & Data Lake, Synapse Analytics, PySpark, Git, LangSmith

Datastores & Visualization: MySQL, MongoDB, Neo4j, Streamlit, Power BI

PROFESSIONAL EXPERIENCE

Indian Political Action Committee (I-PAC) — Hyderabad, India

Lead Associate – AI and Automation

Jun 2025 - Present

News & Social Media Sentiment Tracker AI-Agent (End-to-End GenAI System)

- Built a daily ingestion pipeline orchestrated with **Apache Airflow** and **Docker**, pulling real-time content from social media and news sources through custom web crawlers and third-party APIs. Leveraged parallel task execution in Airflow to process multiple data sources concurrently, reducing end-to-end data processing time and improving pipeline throughput.
- Cleaned and segmented ingested data into aspect-based sentiment categories using a fully local LLM stack (**llama.cpp** + **Mistral 7B**), **LoRA fine-tuned** on custom domain data (**rank = 16, alpha = 32**), eliminating external LLM API dependency while maintaining high classification quality.
- Indexed processed data into **Qdrant** using a hybrid dense + sparse retrieval setup (**Sentence-Transformers** + **BM25**) with metadata tagging, and set up periodic segment-wise **HNSW** reindexing to keep retrieval fast as data volume scaled.
- Designed and deployed a retrieval agent on **LangGraph** and **LangChain**, with query guardrails to block unintended ingestion attempts and exact/semantic caching to cut redundant LLM calls and latency.
- Implemented query translation and decomposition via **prompt engineering** to break complex, multi-intent queries into resolvable sub-queries, routed through an intent-based router across SQL and vector data stores.
- Combined schema-aware prompting with automated retry logic on the SQL side, and hybrid search with cross-encoder reranking on the vector side, to improve retrieval accuracy across both paths.
- Implemented dynamic LLM routing across multiple models based on query complexity to optimize inference cost and latency, while enforcing **100% type-safe structured outputs** using **Pydantic** for all agent responses.
- Built hallucination detection through NLI-based faithfulness checks against retrieved context, paired with output guardrails, and set up continuous **RAG evaluation with RAGAS** plus full pipeline tracing via **LangSmith**.

Senior Associate – AI and Automation

Jul 2022 – May 2025

Social-Media Analytics Wing

- Built an end-to-end **topic modeling pipeline** using **BERTopic** to surface large-scale discussion (~50K records everyday) themes from social media comments, generating sentence embeddings, reducing dimensionality with **PCA**, clustering with **HDBSCAN**, and extracting interpretable, human-readable topic representations via **c-TF-IDF**, eliminating reliance on third-party topic-labeling APIs and lowering associated costs.
- Developed a **Streamlit** dashboard under the SMA (Social Media Analytics) wing to track the performance of party and leader handles across social platforms against defined engagement metrics, and integrated **LangChain** to auto-generate prompt-driven, contextual insights alongside the raw metrics in each report — giving stakeholders a real-time view into digital reach and sentiment while cutting manual reporting effort and reducing associated cost by ~50%, with improved response latency.

- Built a **Selenium**-based web scraping pipeline in Python to extract large-scale election data from public sources, engineering a versioned ingestion layer into **Azure Data Lake** to preserve historical snapshots and ensure data lineage/auditability across election cycles, then processed and transformed the raw data at scale using **PySpark** to produce clean, query-ready datasets that powered downstream **Power BI** dashboards for constituency-level analysis and reporting.

Indian Council of Medical Research (ICMR) – Chennai, India

Biostatistics Intern

Dec 2019 – Feb 2020

- Performed statistical analysis on airborne disease datasets using **R**, and produced analytical reports and visualizations that supported **evidence-based public health insights** and policy-oriented recommendations.

Projects

GraphRAG Chatbot for Football Fans | [Github](#)

- Built a GraphRAG-based chatbot using **Neo4j** to model football entities and relationships (teams, players, matches, tournaments) for context-aware retrieval.
- Integrated LLM-based response generation with structured context injection to reduce hallucinations in entity-heavy QA. Applied **LoRA** fine-tuning / instruction tuning to improve domain relevance and response quality.

LLM From Scratch (PyTorch)

- Built a GPT-style language model from scratch in **PyTorch** as a self-learning exercise to understand transformer internals — implementing core components including multi-head self-attention, positional encoding, layer normalization, and feed-forward blocks, along with tokenization, batched data loading with causal masking, and a full training loop. Evaluated performance via loss curves and text generation samples, exploring sampling strategies like temperature and top-k/top-p.

Multi-Agent System with LangGraph + MCP

- Building a hierarchical multi-agent system in LangGraph, with supervisor agents managing sub-agents to handle complex, multi-step tasks.
- Using **FastMCP** and **MultiServerMCPClient** to connect external tools to the agents in a standard way.

Open-Source Contribution

Swiggy Scraper Python Package | [PyPI](#)

- Developed and published a reusable Python package for Selenium-based Swiggy data extraction with configurable search workflows, **pandas** integration, and PyPI distribution support.
- Implemented production-style packaging using pyproject.toml, wheel builds, semantic versioning, and automated dependency management with uv.

Certifications

Complete Generative AI course with Langchain and HuggingFace. [\(Link\)](#)

Udemy

edX Verified Certificate for Python Basics for Data Science. [\(Link\)](#)

edX

Education

Loyola College, Chennai (M.Sc. Statistics)

Aug 2020 - Jun 2022

Loyola College, Chennai (B.Sc. Statistics)

Jun 2017 - May 2020